BONUS GUIDE — THE POTTERY TEMPLATE VAULT

The Pottery Troubleshooter's Encyclopedia™

47 common problems diagnosed. Causes explained. Fix steps you can apply today.

Cracking · Warping · Glazing · Firing · Surface defects · + Firing Temperature Reference



What's In This Guide

A. Cracking Problems	#1–8 · Before firing, during firing, after firing
B. Warping & Distortion	#9–15 · Bases, rims, tall forms, slabs
C. Glazing Issues	#16–25 · Crawling, pinholes, blistering, color shifts
D. Firing Problems	#26–33 · Explosions, black coring, overfiring, underfiring
E. Structural Failures	#34–40 · Handles, spouts, lids, feet
F. Surface Defects	#41–47 · Bloating, dunting, spalling, contamination
+ Firing Temperature Reference Chart	Cone, °F, °C, and clay body compatibility

> Each entry follows the same format: **What you see → What caused it → How to fix it (step by step)**. When the same problem has multiple possible causes, they are listed from most to least common.

A. Cracking Problems — #1 to #8

1 S-Cracks in the Base

WHAT YOU SEE

An S-shaped crack running through the center of the base after the first firing.

CAUSE

The clay in the base was not compressed enough during throwing. The clay spirals in opposite directions from the center, and when it dries or fires, the tension releases as a crack.

FIX STEPS

→ After opening the floor of the piece, compress the base with your fingertips using firm circular strokes from the center outward.

→ Compress at least twice: once right after opening, once again before you collar the walls.

→ For wider bases (over 6 inches), use a rib tool pressed flat against the base during the final pull to compress the clay fibers inward.

→ Let pieces dry slowly — draping a plastic sheet loosely over them slows drying and reduces stress.

2 Rim Cracks (Small Hairlines at the Top)

WHAT YOU SEE

Thin vertical cracks at the rim, appearing during drying or after bisque.

CAUSE

The rim dried faster than the rest of the piece. Rims have a large surface area-to-volume ratio and lose moisture much faster than the walls or base.

FIX STEPS

→ After throwing, score and compress the rim with a damp sponge or chamois — this compresses the particles and seals the surface slightly.

→ Cover the rim specifically with a thin strip of plastic (a bag cut into strips works well) when slow-drying.

→ Avoid placing pieces near heat sources, fans, or in direct sun during the leather-hard stage.

3 Handle Attachment Cracks

WHAT YOU SEE

A crack at either the top or bottom point where the handle meets the body, appearing after drying or bisque.

CAUSE

Moisture mismatch: the handle and body were at different stages of dryness when joined. As they dry, they shrink at different rates and the join fails.

FIX STEPS

→ Attach handles when both the body and the handle are at the same leather-hard stage — firm enough to hold shape but still flexible.

→ Score both surfaces with a serrated tool, apply slip to both, then press and blend the join with your fingertip.

→ After attaching, wrap the entire piece (including handle) loosely in plastic and dry slowly for 24–48 hours.

→ If attaching a pulled handle (fresh clay) to a leather-hard body: let the pulled handle firm up for 15–30 minutes first, then attach.

4 Thermal Shock Cracks (During or After Firing)

WHAT YOU SEE

A clean, often complete crack through the piece, appearing after the kiln cools or when the piece is moved.

CAUSE

The kiln was cooled too quickly through the quartz inversion temperature range (approximately 1063°F / 573°C), where silica crystals in the clay body contract suddenly.

FIX STEPS

→ Do not open the kiln above 300°F / 150°C — pieces on the top shelf will be hotter.

→ In your kiln schedule, slow the cooling rate between 1200°F and 900°F (650–480°C) to no faster than 100°F/hr (55°C/hr).

→ Electric kilns with digital controllers: program a hold or slow segment in this range.

→ Never place a hot piece on a cold surface or near a window draft.

5 Drying Cracks on Slabs

WHAT YOU SEE

Cracks appearing along the edges or at joins of slab-built pieces during drying.

CAUSE

Uneven drying — slab edges and exposed surfaces dry faster than the center, creating internal stress.

FIX STEPS

→ Dry slabs on plaster bats or between two pieces of canvas — this absorbs moisture evenly from both sides.

→ Flip slabs every few hours during the early drying stage to prevent one side drying faster.

→ When joining slabs, make sure both slabs are at the same moisture level. Score, slip, and blend joins thoroughly.

→ Cover corners and edges with extra plastic during drying — they lose moisture fastest.

6 Cracks at the Base of Tall Forms

WHAT YOU SEE

Horizontal crack running around the piece, about 1–2 inches up from the base, after drying or bisque.

CAUSE

The base dries slower than the walls (it sits on a bat, which holds moisture). The lower walls are pulled as the upper walls try to shrink.

FIX STEPS

→ Use a heat gun or hair dryer (on low) on the base of the bat to encourage even drying.

→ Wire or cut the piece from the bat early — as soon as it's firm enough not to warp — so the base can dry freely.

→ Slow the drying of the upper portion by leaving plastic loosely draped over the top half.

7 Cracks Inside Bowls or Deep Forms

WHAT YOU SEE

Cracks visible inside the bowl, often radiating from the center or running along the floor curve.

CAUSE

Under-compression of the floor during throwing, or a sharp angle at the junction of the floor and wall.

FIX STEPS

→ During throwing, compress the entire interior floor — do not leave the center un-touched.

→ Make the floor-to-wall junction a gentle curve, not a sharp angle. Sharp angles concentrate stress.

→ A rib drawn along the interior from floor to rim helps blend the curve and compress simultaneously.

8 Glaze Craze (Fine Network of Cracks in the Glaze)

WHAT YOU SEE

A web of tiny cracks in the glaze surface only — not through the clay body. Sometimes called "crazing."

CAUSE

The glaze and the clay body contract at different rates during cooling. The glaze shrinks more than the clay beneath it and cracks under tension.

FIX STEPS

→ Adjust the glaze recipe to reduce its silica content slightly (silica expands and contracts significantly).

→ Alternatively, use the same glaze on a different clay body that is a closer thermal match.

→ Slow the cooling rate between 1063°F and 900°F (573–480°C) to reduce thermal shock stress.

→ If the crazing appears only after the piece is used: it may be "delayed crazing" from moisture absorption — apply a second firing to seal the clay body, or switch to a more vitrified clay.

B. Warping & Distortion — #9 to #15

9 Wobbly or Uneven Base

CAUSE

Uneven wall thickness or base thickness; piece placed on an uneven surface while drying; cutting wire pulled unevenly.

FIX STEPS

→ Check wall thickness with a needle tool during throwing. Walls should be consistent from top to bottom.

→ After cutting from the bat, set on a level plaster bat or smooth board.

→ When trimming, check for levelness on the wheel before cutting the foot ring.

10 Rim Warps During Drying

CAUSE

Uneven rim thickness; rim dried faster on one side; piece near a draft source.

FIX STEPS

→ Compress and even the rim with a chamois or rib tool before leaving the wheel.

→ Slow-dry under plastic, away from fans, vents, and windows.

→ For functional ware, a rim template (your templates from the Vault) ensures consistent opening diameter before trimming.

11 Tall Cylinder Collapses or Leans

CAUSE

Walls too thin at the base relative to height; clay too wet or too soft when pulling; over-pulling in a single session.

FIX STEPS

→ Leave the base thicker than the walls — the base carries the weight. A 12-inch cylinder needs a base at least 50% thicker than the walls.

→ If clay feels soft and floppy, use a hair dryer or heat gun (on low) to stiffen the walls before the next pull. Work in sections if the piece is tall.

→ Pull in multiple sessions: throw a short, thick cylinder first, let it stiffen 20–30 minutes under plastic, then continue pulling.

12 Warping During Bisque Firing

CAUSE

Piece not fully dry before loading; kiln fired too fast through the quartz inversion range; piece sitting unevenly in the kiln.

FIX STEPS

→ Bone-dry pieces before loading. If a piece feels even slightly cool to the touch, it still holds moisture — let it dry longer.

→ Include a "candling" segment in your bisque schedule: hold at 200°F/95°C for 1–2 hours to drive off remaining moisture before the temperature rises.

→ Place pieces on a flat kiln shelf. Use kiln wash to prevent sticking, not to correct an uneven shelf.

13 Handles Pull the Body Out of Shape After Firing

CAUSE

Handle too heavy for the wall thickness; handle attached when clay was too soft; asymmetrical handle creating uneven pull during firing.

FIX STEPS

→ Match handle thickness to wall thickness — a handle should not be more than 25% heavier than the wall area where it attaches.

→ Attach handles at the leather-hard stage, not at the soft stage.

→ Pull handles to the same thickness throughout — a thin spot creates an uneven pull.

14 Lids Don't Fit After Firing

CAUSE

Lid and body thrown on different days from different batches of clay; different firing shrinkage; measured incorrectly before drying.

FIX STEPS

→ Always throw the lid and body from the same batch of clay in the same session.

→ Measure both when wet, accounting for the shrinkage rate of your specific clay — not a general average.

→ Fire lids on their bodies when possible (for drop-in lids), or on a kiln shelf with the same orientation.

→ Use your clay's specific shrinkage percentage to calculate the wet measurement needed for a specific fired dimension.

15 Flat Tiles or Plaques Warp

CAUSE

One side drying faster than the other; clay not compressed enough when rolling; drying too quickly.

FIX STEPS

→ Roll slabs for tiles between two pieces of canvas. Flip every few hours as they dry.

→ Place a flat board (same size as the tile) on top during drying to keep them flat.

→ For large tiles, apply compression with a rib in multiple directions after rolling.

→ Fire flat tiles face-down on kiln-washed shelves, or use tile setters (commercially available supports).

C. Glazing Issues — #16 to #25

16 Glaze Crawling (Glaze Pulls Back Into Beads)

CAUSE

Dust, oil, or residue on the bisque surface; glaze applied too thick; glaze has too much plastic clay in the recipe.

FIX STEPS

→ Wipe bisqueware with a damp sponge before glazing to remove dust.

→ Avoid touching the bisque surface with bare hands after cleaning — skin oils repel glaze.

→ Apply glaze in thin, even coats rather than one thick coat.

→ If using a recipe with high kaolin content, reduce it by 5% and substitute silica or feldspar.

17 Pinholes in the Glaze Surface

CAUSE

Gas bubbling through the glaze during firing; pinholes in the clay body releasing; fired too fast through the glaze melt stage.

FIX STEPS

→ Fire more slowly through the final temperature range (last 100°F / 55°C before peak). A longer soak at peak helps bubbles heal over.

→ Ensure bisqueware is fully clean before glazing — dust trapped under glaze becomes gas.

→ If pinholes persist, add a 15-minute hold at peak temperature to your firing schedule.

18 Glaze Color Different From Expected

CAUSE

Different firing temperature, atmosphere, or clay body than the glaze was designed for; glaze applied too thin or too thick.

FIX STEPS

→ Test glazes on a tile from the same clay body before applying to finished pieces.

→ Glaze thickness matters: many glazes fire significantly darker when thin, lighter when thick. Apply 3 even coats for most commercial glazes unless the label says otherwise.

→ If you switched clay bodies, re-test all glazes — the clay's iron content affects oxide-based colors significantly.

19 Glaze Runs Off the Piece

CAUSE

Glaze applied too thick; glaze recipe too fluid at temperature; kiln overfired.

FIX STEPS

→ Leave at least 6mm of unglazed space at the base of the piece.

→ Apply thinner coats and fire a test tile before committing a full piece.

→ If the glaze is fluid by design (e.g., celadon, shino), place pieces on sacrificial tiles with kiln wash to protect the shelf.

20 Glaze Blistering (Bubbles That Don't Heal)

CAUSE

Kiln cooled too fast through the still-molten glaze; organic matter in glaze or clay burning off too late.

FIX STEPS

→ Add a 10-minute soak at peak temperature before beginning cooling.

→ Slow the cooling rate just below peak temperature — this allows the glaze surface to smooth out as it stiffens.

→ Ensure the kiln is vented during early firing to allow organics to escape before the glaze melts and seals the surface.

21 Uneven Glaze Coverage (Bare Spots After Dipping)

CAUSE

Glaze too thick (settles unevenly), bisque too porous or not porous enough, or uneven dipping technique.

FIX STEPS

→ Stir glaze thoroughly before each use — oxides and particles settle quickly.

→ Check glaze specific gravity (use a hydrometer) and add water if too thick.

→ Dip in one smooth motion and hold for 3–5 seconds. Rotate the piece slightly as you withdraw.

→ Touch up thin spots with a brush after the glaze dries.

22 Wax Resist Not Working

CAUSE

Wax emulsion applied too thin; glaze applied before wax fully dried; wax contaminated with water or glaze residue.

FIX STEPS

→ Apply wax in two thin coats, allowing full drying between coats (30+ minutes, longer in humid conditions).

→ Do not apply glaze over wax for at least 1 hour. A simple test: glaze should bead immediately off the surface if the wax is dry.

→ Use a dedicated brush for wax — never share with glaze brushes, as contamination makes wax ineffective.

23 Underglazes Bleed Into Glaze

CAUSE

Glaze applied too thickly over underglaze; underglaze not fully dried before applying clear; underglaze colorants dissolving into the fluid glaze.

FIX STEPS

→ Apply clear glaze thinly over underglazes — 1–2 coats maximum.

→ Let the underglaze dry fully (overnight if needed) before applying clear glaze.

→ Some bright colors (red, orange) are particularly prone to bleeding. Use a manufacturer's recommended clear, not just any clear.

24 Two Glazes Fuse to the Kiln Shelf

CAUSE

Glaze running into contact with the shelf; insufficient kiln wash; piece placed on a shelf with residual glaze from a previous firing.

FIX STEPS

→ Apply kiln wash to every shelf before every glaze firing — including brand new shelves.

→ Leave the bottom 8–10mm of every piece unglazed.

→ After each firing, scrape and re-apply kiln wash to any area where glaze touched the shelf.

→ For fluid glazes, place pieces on a sacrificial tile (same clay body, kiln-washed) that can be replaced if the glaze runs.

25 Metallic Sheen or Matte Spots on a Glossy Glaze

CAUSE

Reduction atmosphere in an electric kiln (uncommon); glaze formulation issue; piece placed too close to the element.

FIX STEPS

→ Ensure adequate kiln ventilation throughout the firing — trapped gases cause localized atmosphere changes.

→ Keep pieces at least 1 inch from the kiln elements and from the kiln walls.

→ If the problem is consistent, check the glaze recipe for zinc oxide (which can go matte under reduction conditions).

D. Firing Problems — #26 to #33

26 Piece Explodes in the Kiln

CAUSE

Trapped air pockets in the clay body; piece not fully dry before firing; firing too fast through the steam-release stage (below 212°F/100°C).

FIX STEPS

→ Wedge clay thoroughly until no air pockets remain — check by cutting with a wire tool and looking for bubbles.

→ Ensure pieces are bone-dry before loading. If in doubt, put a piece against your cheek — it should feel room-temperature, not cool.

→ Include a candling hold at 200°F/95°C for 1–2 hours at the start of every bisque schedule.

27 Black Core in Thick Pieces After Firing

CAUSE

Carbon trapped inside thick clay walls because the kiln fired too fast — organics did not fully burn out before the surface sealed.

FIX STEPS

→ Slow the firing rate between 700°F and 1400°F (370–760°C) to allow complete carbon burnout.

→ Vent the kiln during this range to provide oxygen. In electric kilns, prop the lid or leave the peepholes open.

→ For thick pieces (over 3/4 inch), extend the hold at bisque temperature by 15–30 minutes.

28 Underfired Glaze (Still Rough or Matte When Glossy Expected)

CAUSE

Kiln did not reach target temperature; elements aging and losing efficiency; thermocouple drift.

FIX STEPS

→ Use witness cones (or cone packs) in every load — do not rely solely on the kiln's digital readout.

→ Test the kiln's actual temperature by placing cone packs at the top, middle, and bottom of the chamber.

→ If elements are more than 2 years old and the kiln is struggling to reach temperature, replace them.

29 Overfired Pieces (Bloated, Slumped, Darkened)

CAUSE

Kiln exceeded target temperature; witness cones not used; thermocouple reading lower than actual kiln temperature.

FIX STEPS

→ Always use witness cones. A digital controller is a convenience, not a guarantee.

→ Check and replace your kiln's thermocouple every 1–2 years (it drifts over time).

→ After any kiln repair or element replacement, fire a test load of sacrificial pieces before firing finished work.

30 Kiln Shelves Crack

CAUSE

Thermal shock from cooling too fast; glaze dripping onto the shelf surface causing stress.

FIX STEPS

→ Cool kilns slowly — never force-cool by opening above 300°F/150°C.

→ Keep shelves kiln-washed and free of glaze drips. Refire glaze-contaminated shelves on their own (face down) to burn off residue.

→ Store kiln shelves lying flat — not standing on edge — to prevent warping.

31 Pieces Stick to Each Other in the Kiln

CAUSE

Pieces placed too close together; glaze on rims or sides touching during firing; glaze running and bridging the gap.

FIX STEPS

→ Maintain at least 1/2 inch between pieces at all times.

→ Do not glaze rims where two pieces might touch. If stacking (e.g., bowls), use a kiln wash-covered saddle between them.

→ Never glaze the interior of a piece if the interior will touch another piece during firing.

32 Uneven Color Across a Single Piece

CAUSE

Kiln temperature variation between top and bottom shelves; piece placed near the element; uneven glaze application.

FIX STEPS

→ Cone packs at different shelf levels will show temperature variations. Place consistent-sized pieces together to minimize variation.

→ Rotate the position of pieces (top shelf to bottom shelf) between firings if you fire the same glazes repeatedly.

33 Smoke or Fumes From the Kiln

CAUSE

Organic matter in clay or glaze burning off; paper clay components carbonizing; wax resist burning before the firing temperature rises.

FIX STEPS

→ Fire kilns only in a ventilated space. Do not close garage doors, studio doors, or windows during firing.

→ During the 400–1400°F (200–760°C) range, vent the kiln to release gases. This is normal and expected, not a sign of a problem.

→ If smoke is heavy or unusual, stop the firing, allow the kiln to cool, and identify the source before refiring.

E. Structural Failures — #34 to #40

34 Handle Falls Off During Firing

CAUSE

Poor join — insufficient scoring and slipping; moisture mismatch at attachment; thin walls at attachment point.

FIX STEPS

→ Score both the body and the handle end in a crosshatch pattern with a serrated rib or fork before applying slip.

→ Use slip made from the same clay body, not water — water weakens the join.

→ Press and blend the join so there is no visible seam — the handle should look grown from the body, not applied to it.

35 Lid Knob Breaks Off

CAUSE

Knob attached when body was too dry; insufficient scoring; knob too thin at the base relative to its height.

FIX STEPS

→ Attach knobs only when the lid is at the same leather-hard stage. If the lid has dried more, score the lid surface and rehydrate slightly with a damp brush before attaching.

→ Knob base diameter should be at least 1/3 the knob height for stability.

→ For functional ware, always attach thrown knobs rather than coil knobs — they have a stronger attachment structure.

36 Spout Cracks at the Join

CAUSE

Spout and body at different moisture levels; spout too heavy for the wall thickness; sharp angle at the join point.

FIX STEPS

→ Dry the spout and the body to the same leather-hard stage before joining.

→ Blend the join smoothly from both inside and outside where accessible.

→ Support the spout with a small wad of newspaper or foam under it during drying to prevent it from pulling down with gravity.

37 Foot Ring Cracks or Pops Off

CAUSE

Trimming too deep and breaking through the base; foot ring too narrow; uneven wall thickness around the foot area.

FIX STEPS

→ Leave the base at least 4–5mm thick at the foot ring center when trimming.

→ The foot ring width should be proportional to the piece — a wide bowl needs a wide foot ring for stability and to avoid cracking during firing shrinkage.

→ Use a needle tool to probe base depth during trimming if you are unsure how much clay remains.

38 Coil Joins Open Up During Firing

CAUSE

Coils not blended thoroughly enough; coils applied when previous layer was too dry; inadequate compression of the join.

FIX STEPS

→ Blend coils on both the interior and exterior surfaces until no seam line is visible.

→ Score the previous layer and add fresh slip before attaching each new coil.

→ Work on the interior while you can still reach — once the piece narrows at the top, interior blending is impossible.

39 Piece Cracks at a Thick-to-Thin Wall Transition

CAUSE

The thick section dries slower and shrinks at a different rate than the thin section, causing stress at the boundary.

FIX STEPS

→ Maintain consistent wall thickness throughout the piece. Use a wall thickness gauge or needle tool to check during throwing.

→ If a thick-to-thin transition is unavoidable (e.g., at the base of a handle), dry the piece extra slowly and wrap the transition area in plastic during early drying.

40 Piece Loses Its Shape Between Throwing and Firing

CAUSE

Clay too soft when removed from the wheel; inadequate support during drying; disturbed before leather-hard stage.

FIX STEPS

→ Allow pieces to stiffen on the bat for at least 20–30 minutes before moving, longer for large pieces.

→ Support the shape with crumpled newspaper or foam if the piece is prone to collapsing (e.g., open bowls, oval forms).

→ Move pieces by lifting from the base, never from the rim or walls.

F. Surface Defects — #41 to #47

41 Bloating (Large Bubbles Under the Glaze)

CAUSE

Over-vitrified clay body trapping gases; organic matter or contamination inside the clay; overfired.

FIX STEPS

→ Check that bisque temperature is correct for your clay body — over-bisquing seals the surface before organics escape.

→ Do not overfire glaze — when the glaze is fluid, it seals the surface and traps any remaining gas bubbles.

→ Inspect clay for contamination (plaster, organic debris) before using. White specks can be plaster, which causes spalling.

42 Plaster Contamination — Spalling Spots

CAUSE

Tiny plaster fragments in the clay body expand during firing and pop off the surface, creating a pit or crater.

FIX STEPS

→ Never allow plaster tools, plaster bats, or plaster to contact wet clay that you intend to fire — plaster fragments are impossible to remove once embedded.

→ Use only wood, metal, plastic, or canvas tools and bats for clay that will be fired.

→ If you see white specks in reclaimed clay, test-fire a small piece before using the whole batch. If spots appear, discard the batch.

43 Lime Popping (Pits in Fired Earthenware)

CAUSE

Calcium carbonate (limestone) in the clay body expands after firing when it absorbs moisture from the air, popping off the surface.

FIX STEPS

→ Fire earthenware to a high enough temperature to fully convert calcium carbonate to calcium silicate — check your clay body's specification sheet.

→ Immediately after firing, store unfired earthenware away from moisture until sealed with glaze.

→ For outdoor earthenware, use fully vitrified clay bodies that do not absorb moisture.

44 Mold or Efflorescence (White Powder on Fired Surface)

CAUSE

Soluble salts from the clay migrating to the surface during drying; fired but unglazed surface absorbing moisture from the environment.

FIX STEPS

→ Use a barium carbonate wash on the exterior of greenware before bisque firing — it neutralizes soluble salts.

→ Fire functional wares with a watertight glaze on all interior surfaces.

→ Store unglazed bisqueware in dry conditions.

45 Glaze Flaking Off After Use

CAUSE

Poor glaze-to-clay fit (compression or tension mismatch); glaze in compression spalls off over time, especially after dishwasher cycles.

FIX STEPS

→ Test all glazes for fit using the freeze-thaw test: fire a test tile, fill it with water, and freeze overnight 3 times. Flaking indicates poor fit.

→ Adjust the glaze recipe to reduce the mismatch — a glaze formulation specialist or your clay manufacturer can advise on compatible glazes for your clay body.

46 Dark Spots or Flashing on the Surface

CAUSE

Iron or manganese in the clay body migrating to the surface during firing; proximity to other pieces or kiln elements causing localized atmosphere changes.

FIX STEPS

→ If the spots are consistent and you are using an iron-bearing clay, this may be a feature rather than a defect — test your market's response before changing the clay.

→ If undesired, switch to a low-iron white stoneware body for consistent, predictable results.

→ Ensure pieces are not too close to elements or other pieces — localized hot spots cause flashing.

47 Scratches on the Glazed Surface After Use

CAUSE

Glaze fired too soft (underfired); glaze formulation lacks sufficient silica hardness; cutlery or abrasive cleaning.

FIX STEPS

→ Fire to the correct cone for your glaze — matte glazes fired to their minimum cone scratch easily.

→ Test hardness by running a knife over a test tile — functional glaze should not scratch. If it does, fire higher or adjust the recipe.

→ Advise customers to hand-wash pieces with soft cloths and avoid stacking.

Firing Temperature Reference Chart

Use witness cones for every firing — this chart shows the target temperature for each cone, the typical clay bodies used at each range, and what to expect from your glaze and clay at each temperature.

Cone	°F (approx.)	°C (approx.)	Range Name	Compatible Clay Bodies	Notes
022	1087°F	586°C	Bisque / Low Fire	Earthenware, red art clay	Bisque only — very soft. Lusters and overglaze enamels.
06	1828°F	998°C	Low Fire	Earthenware, terracotta	Standard bisque for most earthenwares. Non-vitrified.
05	1888°F	1031°C	Low Fire	Earthenware, underglaze work	Common underglaze firing temperature. Still porous.
04	1945°F	1063°C	Low Fire	Earthenware, red stoneware	Standard earthenware glaze firing. Bright, matte, and satin glazes.
02	2048°F	1120°C	Low-Mid Fire	Earthenware (high), semi-vitreous	Upper range of commercial low-fire glazes.
1	2109°F	1154°C	Mid Fire	White stoneware, B-mix	Lower mid-range. Some stone body begins to vitrify.
5	2167°F	1186°C	Mid Fire	Stoneware, B-mix, porcelain	Beginning of mid-fire range. Many commercial studio glazes designed for C/5-C/6.
6	2232°F	1222°C	Mid Fire	Stoneware, porcelain, B-mix	Most common studio glaze temperature. Vitrified, food-safe

					when properly glazed.
8	2305°F	1263°C	High Fire	Stoneware, high-fire porcelain	Upper stoneware range. Glazes flow more readily.
10	2381°F	1305°C	High Fire	High-fire stoneware, hard porcelain	Traditional reduction stoneware range. Very vitrified, dense, non-porous.
11	2399°F	1315°C	High Fire	Hard-paste porcelain	Maximum for most studio kilns. Traditional East Asian porcelain range.

Important: These temperatures are approximate. Kiln atmosphere (oxidation vs. reduction), firing speed, kiln size, and element age all affect the actual temperature your clay and glaze experience. Always use witness cones — they measure heat work (time × temperature combined), not just temperature, which is what actually matters to your clay and glaze.

Next Step

When a piece fails, check Section A first (cracking is the most common). Use this guide to identify the cause before making your next batch — catching the root issue saves the next piece, not just this one.